

Total DOF: $23 + 2 - 2 - ((2-1)(2-1)) - 2 - 16 = 0$

mixer

MW CO = 28 g/mol

$\frac{45000 \text{ kg}}{\text{hr}} \cdot \frac{31.2 \text{ g/mol}}{\text{mol}} = 14 \text{ g}$

$M = 0.5 \text{ mole} \cdot \frac{28 \text{ g}}{\text{mol}} = 14 \text{ g}$

MW CO₂ = 44 g/mol

$\frac{31.2 \text{ g/mol}}{\text{mol}} \cdot \frac{1742.3 \text{ kmol}}{\text{hr}} = 8.8 \text{ g}$

$D = 0.2 \text{ mol} \cdot \frac{44 \text{ g}}{\text{mol}} = 8.8 \text{ g}$

MW N₂ = 28 g/mol

$\frac{1742.3 \text{ kmol}}{\text{hr}} \cdot \frac{28 \text{ g}}{\text{mol}} = 8.4 \text{ g}$

$N = 0.3 \cdot \frac{28 \text{ g}}{\text{mol}} = 8.4 \text{ g}$

MW W = 18 g/mol

MW CH₃COOH = 60 g/mol

$m_1 = 721.5 \frac{\text{kmol}}{\text{hr}}$

$\rightarrow m_3 = m_1 = 721.5 \frac{\text{kmol}}{\text{hr}}$

$W_3 = W_2 + W_9$

$D_1 = 288.46 \frac{\text{kmol}}{\text{hr}}$

$D_3 = D_1 = 288.46 \frac{\text{kmol}}{\text{hr}}$

$A_3 = A_9 = 26.62 \frac{\text{kmol}}{\text{hr}}$

$N_1 = 432.69 \frac{\text{kmol}}{\text{hr}}$

$N_3 = N_1 = 432.69 \frac{\text{kmol}}{\text{hr}}$

reactor

$m_4 = m_3 - 6\dot{E}_1 - 4\dot{E}_2 = 0.1557m_3 \rightarrow 721.5 - 6\dot{E}_1 - 4\dot{E}_2 = 0.1557(721.5) = 112.28 \frac{\text{kmol}}{\text{hr}}$

$D_4 = D_3 + 4\dot{E}_1 + 2\dot{E}_2 \rightarrow 288.46 + 4\dot{E}_1 + 2\dot{E}_2 = 288.46 + 4(86.96) + 2(21.78) = 674.86$

$N_4 = N_3 = 432.69 \frac{\text{kmol}}{\text{hr}}$

$A_5 = A_3 + \dot{E}_2 = A_9 + \dot{E}_2 \rightarrow A_9 = 48.4 - 21.78 = 26.62 \frac{\text{kmol}}{\text{hr}}$

$E_5 = \dot{E}_1 = 86.96 \frac{\text{kmol}}{\text{hr}}$

$W_5 = W_3 - 3\dot{E}_1 - 2\dot{E}_2 \rightarrow W_2 + W_9 - 3\dot{E}_1 - 2\dot{E}_2$

$m_4 = 112.28 \frac{\text{kmol}}{\text{hr}} \cdot 721.5 \frac{\text{kmol}}{\text{hr}} - 6(86.96 \frac{\text{kmol}}{\text{hr}}) - 4\dot{E}_2 \Rightarrow \dot{E}_2 = 21.78 \frac{\text{kmol}}{\text{hr}}$

separator

$W_5 = W_6 + W_7$

$E_6 = 0.125 \cdot 32000 \frac{\text{kg}}{\text{hr}} \cdot \frac{\text{mol}}{46 \text{ g}} = 86.96 \frac{\text{kmol}}{\text{hr}}$

$E_5 = E_6 = 86.96 \frac{\text{kmol}}{\text{hr}}$

$W_8 = 0.876(32000 \frac{\text{kg}}{\text{hr}}) \left(\frac{\text{mol}}{18 \text{ g}} \right) = 1555.56 \frac{\text{kmol}}{\text{hr}}$

$A_5 = A_7 = 48.9 \frac{\text{kmol}}{\text{hr}}$

$n_6 = 1555.56 \frac{\text{kmol}}{\text{hr}} + 86.96 = 1642.51 \frac{\text{kmol}}{\text{hr}}$

overall

$$M_H = M_1 - 6E_1 - 4E_2$$

$$D_4 = D_1 - 4E_1 + 2E_2$$

$$N_1 = N_4$$

$$3E_1 + 2E_2 = W_2 - W_6 - W_8 \rightarrow W_2 = 3E_1 + 2E_2 + W_6 + W_8$$

$$A_8 = E_2 = 21.78 \frac{\text{kmol}}{\text{h}} = 0.45 A_2 \Rightarrow A_2 = 48.4 \frac{\text{kmol}}{\text{h}}$$

$$\dot{n}_2 = W_2 = W_6 + W_8 + 304.44 \rightarrow 1555.56 + 3448.43 + 304.44 = 5308.43 \checkmark$$

$$W_3 = W_2 + W_9 = 5308.43 + 4214.75 = 9523.18 \frac{\text{kmol}}{\text{h}} \checkmark$$

$$W_5 = 0.96 m_5 \cdot \frac{\text{mol}}{18g} = 9218.75 \frac{\text{kmol}}{\text{h}}$$

$$6904.16 + \left(\frac{368.75}{0.04} \right) = 9218.75 \checkmark$$

$$W_7 = W_5 - 1555.56 = 9218.75 - 1555.56 = 7663.19 \checkmark$$

$$W_8 = 0.45(7663.19) = 3448.43 \checkmark$$

$$W_4 = 0.55 W_7 = 0.55(7663.19) = 4214.75 \checkmark$$

$$\text{Stream 1} \rightarrow \dot{n}_1 = 1442.3 \frac{\text{kmol}}{\text{h}}$$

$$m_1 = 721.15 \frac{\text{kmol}}{\text{h}}$$

$$D_1 = 288.96 \frac{\text{kmol}}{\text{h}}$$

$$N_1 = 432.69 \frac{\text{kmol}}{\text{h}}$$

$$\text{Stream 2} \rightarrow \dot{n}_2 = W_2 = 5308.43 \frac{\text{kmol}}{\text{h}}$$

$$\text{Stream 4} \rightarrow \dot{n}_4 = 1229.83 \frac{\text{kmol}}{\text{h}}$$

$$m_{H_2} = 112.28 \frac{\text{kmol}}{\text{h}}$$

$$D_4 = 679.86 \frac{\text{kmol}}{\text{h}}$$

$$N_4 = 432.69 \frac{\text{kmol}}{\text{h}}$$

$$\text{Stream 3} \rightarrow \dot{n}_3 = 1099.21$$

$$m_3 = 721.15 \frac{\text{kmol}}{\text{h}}$$

$$D_3 = 288.96 \frac{\text{kmol}}{\text{h}}$$

$$N_3 = 432.69 \frac{\text{kmol}}{\text{h}}$$

$$A_3 = 26.62 \frac{\text{kmol}}{\text{h}}$$

$$W_3 = 9523.18 \frac{\text{kmol}}{\text{h}}$$

$$\text{Stream 5} \rightarrow \dot{n}_5 = 9354.11 \frac{\text{kmol}}{\text{h}}$$

$$W_5 = 9218.75 \frac{\text{kmol}}{\text{h}}$$

$$E_5 = 86.96 \frac{\text{kmol}}{\text{h}}$$

$$A_5 = 48.9 \frac{\text{kmol}}{\text{h}}$$

Stream 6 $\rightarrow \dot{n}_6 = 1642.52 \text{ kmol/hr}$
 $\dot{W}_6 = 1555.56 \text{ kmol/hr}$
 $\dot{E}_6 = 86.96 \text{ kmol/hr}$

Stream 7 $\rightarrow \dot{n}_7 = 711.59 \text{ kmol/hr}$
 $\dot{W}_7 = 7668.19 \text{ kmol/hr}$
 $\dot{A}_7 = 48.4 \text{ kmol/hr}$

Stream 8 $\rightarrow \dot{n}_8 = 3470.71 \text{ kmol/hr}$
 $\dot{A}_8 = 21.78 \text{ kmol/hr}$
 $\dot{W}_8 = 3948.43 \text{ kmol/hr}$

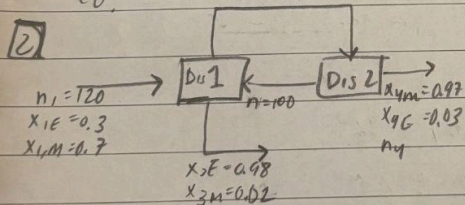
Stream 9 $\rightarrow \dot{n}_9 = 4241.87 \text{ kmol/hr}$
 $\dot{A}_9 = 26.62 \text{ kmol/hr}$
 $\dot{W}_9 = 4219.85 \text{ kmol/hr}$

single pass

conversion $= \frac{(9523.18 \text{ kmol/hr} - 9218.75 \text{ kmol/hr})}{9523.18} \cdot 100 = 3.19\%$

overall $= \frac{5308.43 - (1555.56 + 3948.48)}{5308.43} = 5.73\%$

to improve the efficient use of water there could be more CO₂



Dis 1

E: $0.1n_2 + 0.98n_3 = 0.3100 + x_{SE} \cdot 100 \rightarrow x_{SE} = \frac{0.1 \cdot 122.6 + 0.98 \cdot 224 - 30}{100} = 0.14$

M: $0.9n_2 + 0.02n_3 = 0.2100 + x_{SM} \cdot 100 \rightarrow x_{SM} = 0.86$

$n_2 + n_3 = n_1 + n_5 \rightarrow n_2 = 100 + 100 - 27.4 = 172.6 \frac{\text{kmol}}{\text{hr}}$

Dis 2

E: $0.3n_4 + x_{SE} \cdot 100 = 0.1n_2$

E: $0.03n_4 + 0.98n_3 = 0.3102$

M: $0.9n_4 + x_{SM} \cdot 100 = 0.9n_2$

M: $0.17n_4 + 0.22n_3 = 67.1$

$n_3 = \frac{70 - 0.97n_4}{0.02} = 27.4 \frac{\text{kmol}}{\text{hr}}$

$0.03n_4 = 30 - 0.98 \cdot \left(\frac{70 - 0.97n_4}{0.02} \right) \rightarrow 0.03n_4 = 30 - 3436 = 47.53n_4 \rightarrow n_4 = 71.6$

20/35

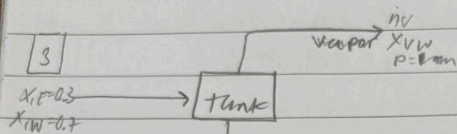
Stream 4

$$f_{RUE} = \frac{0.3 \cdot 71.6}{0.1 + 12.6} = 1.29 \quad f_{RUM} = \frac{0.2 \cdot 71.6}{0.7 + 22.6} = 0.3$$

$$f_{R35} = \frac{0.98(22.4)}{(0.3 \cdot 100) + (0.14 \cdot 100)} = 0.61$$

$$f_{RSM} = \frac{0.02(22.4)}{(0.1 + 100) + (0.86 \cdot 100)} = 0.0035$$

-5 pts
no recycle?
Stream?
comps?
-10 pts



basis $n_F = 100$

$$F = 0.3 \cdot 100 = X_F n_V + X_F n_L$$

$$W = 0.2 \cdot 100 = X_W n_V + X_W n_L$$

$$K_W = K_F = 10^A - 10^{\frac{B}{T+C}}$$

$$0 = 0.3 \cdot \frac{10^{\frac{8.04}{760}} - 10^{\frac{1554.3}{7+222.65}}}{7+0.5 \left(\frac{10^{\frac{8.04}{760}} - 10^{\frac{1554.3}{7+222.65}}}{760} \right) - 1} + 0.7 \left(\frac{10^{\frac{1330.22}{760}} - 10^{\frac{1330.22}{7+225}}}{7+0.5 \left(\frac{10^{\frac{1330.22}{760}} - 10^{\frac{1330.22}{7+225}}}{760} \right) - 1} \right)$$

$$T = 93.39^\circ\text{C}$$

$$K_F = 10^{\frac{8.04}{760} - \frac{1554.3}{7+222.65}} = 9.74$$

$$K_W = 10^{\frac{8.04}{760} - \frac{1554.3}{7+222.65}} = 0.79$$

$$X_{VF} = \frac{0.3}{1+0.5(1.75-1)} = 0.22$$

$$X_{VW} = \frac{0.7}{1+0.5(1.75-1)} = 0.78$$

$$X_{LF} = K_F \cdot X_{VF} = 0.39$$

$$X_{LW} = K_W \cdot X_{VW} = 0.61$$

$$n_V = \frac{30 - 0.34 n_L}{0.22}$$

$$0.61 n_L = 70 - 0.78 \left(\frac{30 - 0.34 n_L}{0.22} \right)$$

$$0.61 n_L = 70 - 106 + 1.38 n_L$$

$$n_L = 46.25 \quad n_V = 53.48$$

$$f_{REV} = \frac{0.22 \cdot 53.48}{30} = 0.39$$

$$f_{RUV} = \frac{0.78 \cdot 53.48}{70} = 0.601$$

$$f_{REL} = \frac{0.39 \cdot 46.25}{30} = 0.6$$

$$f_{RUL} = \frac{0.61 \cdot 46.25}{70} = 0.4$$

-2 pts
no sep. Factor?
-5 pts